UBER TRIPS ANALYTICS

From raw trip records to clear business decisions

SQL ANALYSIS • 10 QUESTIONS • 2017–2020

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SQL • OPERATIONS • REVENUE • DEMAND



NYC GREEN TAXI TRIP ANALYSIS

SQL • Demand • Revenue • Geography

SQL-only presentation
Power BI deck will be delivered separately

How to read this deck

EVERY QUESTION INCLUDES THE QUESTION, SUBMITTED SQL SCREENSHOT, RESULT SCREENSHOT, AND INTERPRETATION.

01–05	Trips, revenue, peak times, borough share, frequent routes
06–10	Passenger load, trip distance, payment type, tips, borough-pair revenue
Evidence	Uploaded SQL screenshots are included as supplied; result tables are interpreted in plain language
Quality	Potential query/data issues are called out rather than hidden

Note: SQL screenshots are placed as images for faithful visibility; the text can be edited by replacing the screenshot.

Question 01 | Yearly Trip Trends

CALCULATE TOTAL TRIPS PER YEAR AND YEAR-OVER-YEAR (YOY) PERCENTAGE CHANGE.

BUSINESS QUESTION 01 • SQL QUERY

```sql
# 1. Yearly Trip Trends: Calculate total trips per year and percentage change year-over-year.
WITH yearly_trips AS (
    SELECT 2017 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2017
UNION ALL
SELECT 2018 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2018
UNION ALL
SELECT 2019 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2019
    UNION ALL
SELECT 2020 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2020
),
yearly_comparison AS (
SELECT trip_year,total_trips,LAG(total_trips) OVER (ORDER BY trip_year) AS previous_year_trips FROM yearly_trips
)SELECT trip_year, total_trips, previous_year_trips,ROUND((total_trips - previous_year_trips)* 100.0 / NULLIF(previous_year_trips, 0),2
    ) AS yoy_percentage_change
FROM yearly_comparison
ORDER BY trip_year;
```

Question 01 | Result & interpretation

YEARLY TRIP TRENDS • OUTPUT PROVIDED IN YOUR SCREENSHOTS

Result Grid | Filter Rows: | Export: | Wrap Cell Co

trip_year	total_trips	previous_year_trips	yoy_percentage_change
2017	1048575	NULL	NULL
2018	8807303	1048575	1048575
2019	6045506	8807303	-31.36
2020	1735507	6045506	-71.29

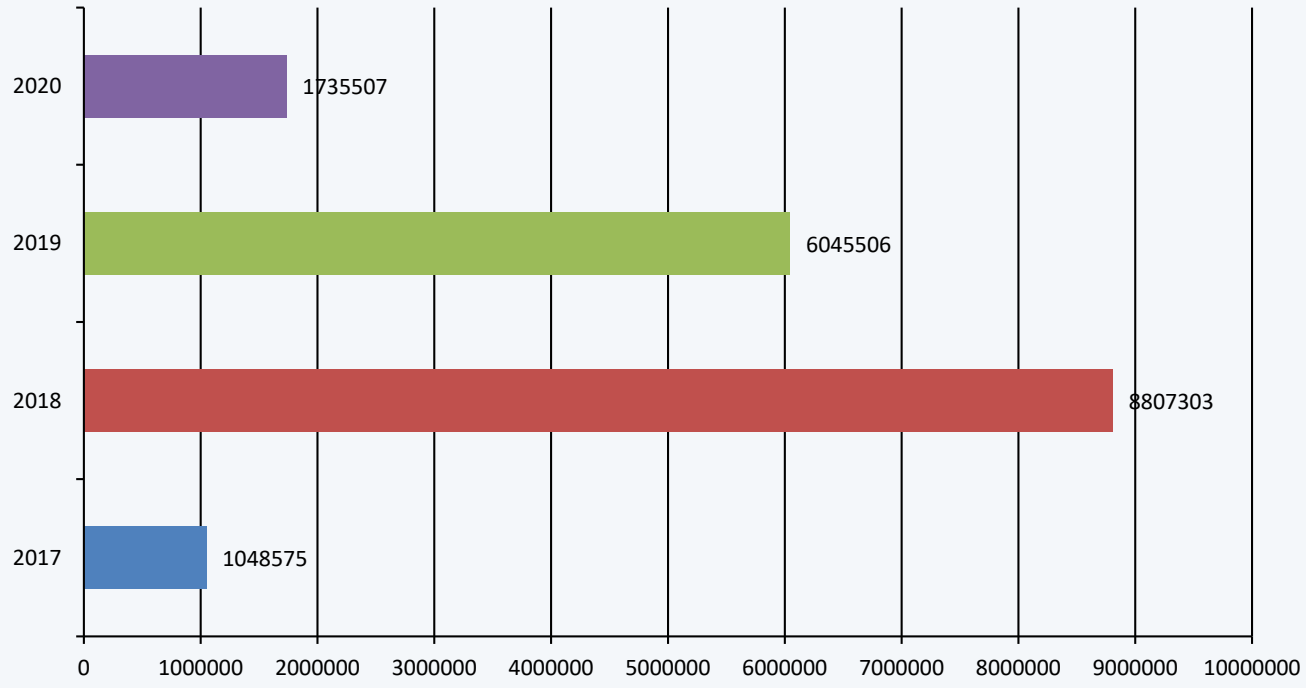
KEY TAKEAWAY

2018 has the highest trip count in the supplied output (8,807,303). Trips then decline by 31.36% in 2019 and 71.29% in 2020. The sharp jump from 2017 to 2018 should be validated against source-table coverage.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 01 | Yearly Trip Trends

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



The supplied output peaks in 2018, then falls sharply in 2019 and 2020.

WHAT THE RESULT TELLS US

- 2018 records the highest trip count in this result set.
- 2019 is down about 31.4% from 2018; 2020 is down about 71.3% from 2019.
- Validate year coverage and completeness before interpreting the drop as a real demand change.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 02 | Monthly Revenue Insights

CALCULATE MONTHLY TOTAL REVENUE AND AVERAGE REVENUE PER TRIP.

BUSINESS QUESTION 02 • SQL QUERY

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

trip_year	trip_month	month_name	total_revenue	average_revenue_per_trip
2017	1	January	4216.61	14.69
2017	2	February	4037.88	13.03
2017	3	March	4088.43	13.27
2017	4	April	3469.07	13.55
2017	5	May	3718.88	13.23
2017	6	June	4357.57	13.75
2017	7	July	3688	14.02
2017	8	August	4323	14.6
2017	9	September	3734.25	12.92
2017	10	October	4300.22	13.35
2017	11	November	3696.56	14.11
2017	12	December	4673.08	13.47
2018	1	January	11115132.16	14
2018	2	February	10901751.63	14.16
2018	3	March	12481048.03	14.91

Question 02 | Result & interpretation

MONTHLY REVENUE INSIGHTS • OUTPUT PROVIDED IN YOUR SCREENSHOTS

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

trip_year	trip_month	month_name	total_revenue	average_revenue_per_trip
2017	1	January	4216.61	14.69
2017	2	February	4037.88	13.03
2017	3	March	4088.43	13.27
2017	4	April	3469.07	13.55
2017	5	May	3718.88	13.23
2017	6	June	4357.57	13.75
2017	7	July	3688	14.02
2017	8	August	4323	14.6
2017	9	September	3734.25	12.92
2017	10	October	4300.22	13.35
2017	11	November	3696.56	14.11
2017	12	December	4673.08	13.47
2018	1	January	11115132.16	14
2018	2	February	10901751.63	14.16
2018	3	March	12481048.03	14.91

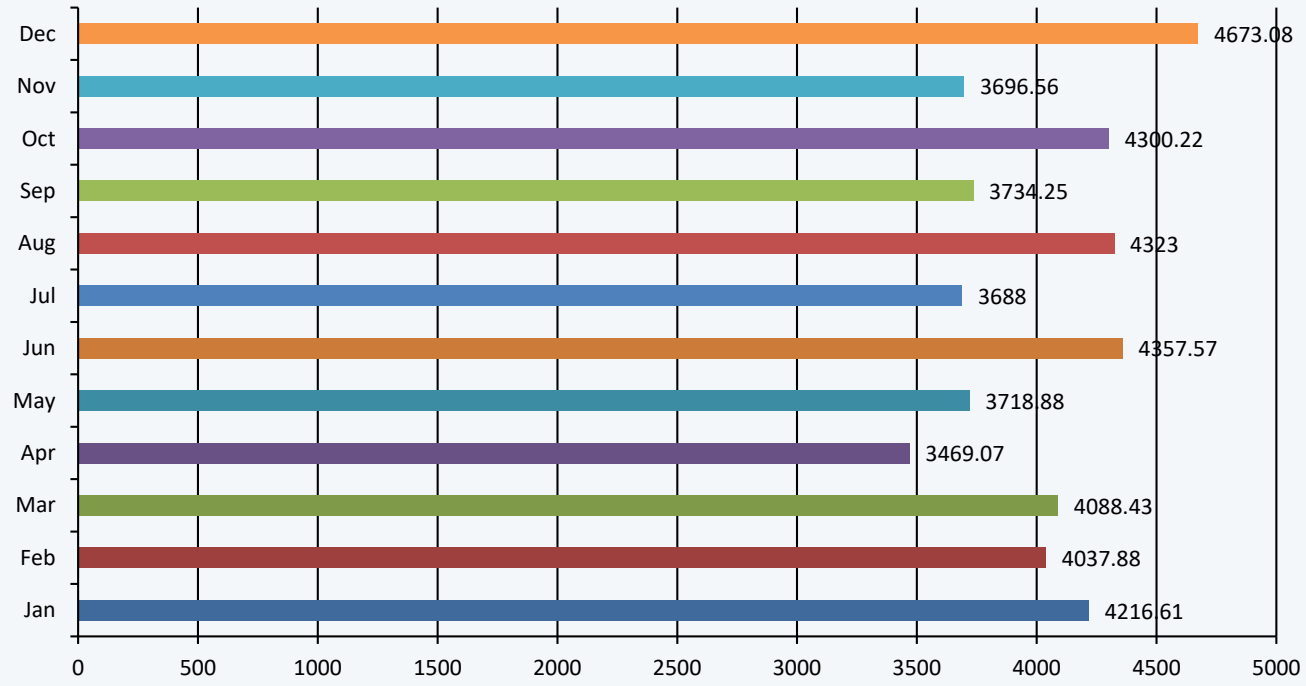
KEY TAKEAWAY

The displayed 2017 monthly totals range from about 3,469 to 4,673, with December highest among the shown months. The 2018 values are in the millions, so verify units, filters and table coverage before comparing years.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 02 | Monthly Revenue Insights

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



December 2017 is the highest of the visible 2017 months; cross-year totals need validation.

WHAT THE RESULT TELLS US

- Visible 2017 monthly revenue is roughly 3.47K–4.67K in the screenshot.
- December is highest among the displayed 2017 months.
- 2018 values shown are much larger; confirm units, filters, and source-table consistency.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 03 | Peak Day & Time Analysis

IDENTIFY THE DAY OF WEEK AND HOUR WITH THE HIGHEST AVERAGE TRIP VOLUME.

BUSINESS QUESTION 03 • SQL QUERY

SQL logic (reconstructed from the supplied result screenshot)

```
WITH all_trips AS (
  SELECT lpep_pickup_datetime AS pickup_date FROM stg_uber_2017
  UNION ALL SELECT lpep_pickup_datetime FROM stg_uber_2018
  UNION ALL SELECT lpep_pickup_datetime FROM stg_uber_2019
  UNION ALL SELECT lpep_pickup_datetime FROM stg_uber_2020
), daily_hourly_trips AS (
  SELECT DATE(pickup_date) AS trip_date,
        DAYOFWEEK(pickup_date) AS day_num,
        HOUR(pickup_date) AS trip_hour, COUNT(*) AS total_trips
  FROM all_trips WHERE pickup_date IS NOT NULL
  GROUP BY DATE(pickup_date), DAYOFWEEK(pickup_date), HOUR(pickup_date)
), average_trips AS (
  SELECT day_num, trip_hour, ROUND(AVG(total_trips),2) AS avg_trips
  FROM daily_hourly_trips GROUP BY day_num, trip_hour
)
SELECT day_num, trip_hour, avg_trips FROM average_trips
ORDER BY avg_trips DESC LIMIT 10;
```

This is a readable reconstruction, not a screenshot of the original query.

Question 03 | Result & interpretation

PEAK DAY & TIME ANALYSIS • OUTPUT PROVIDED IN YOUR SCREENSHOTS

day_of_week	trip_hour	time_slot	avg_trips
Friday	17	17:00 - 18:00	1166.86
Friday	19	19:00 - 20:00	1120.03
Thursday	18	18:00 - 19:00	1115.69
Friday	16	16:00 - 17:00	1086.65
Thursday	17	17:00 - 18:00	1085.39

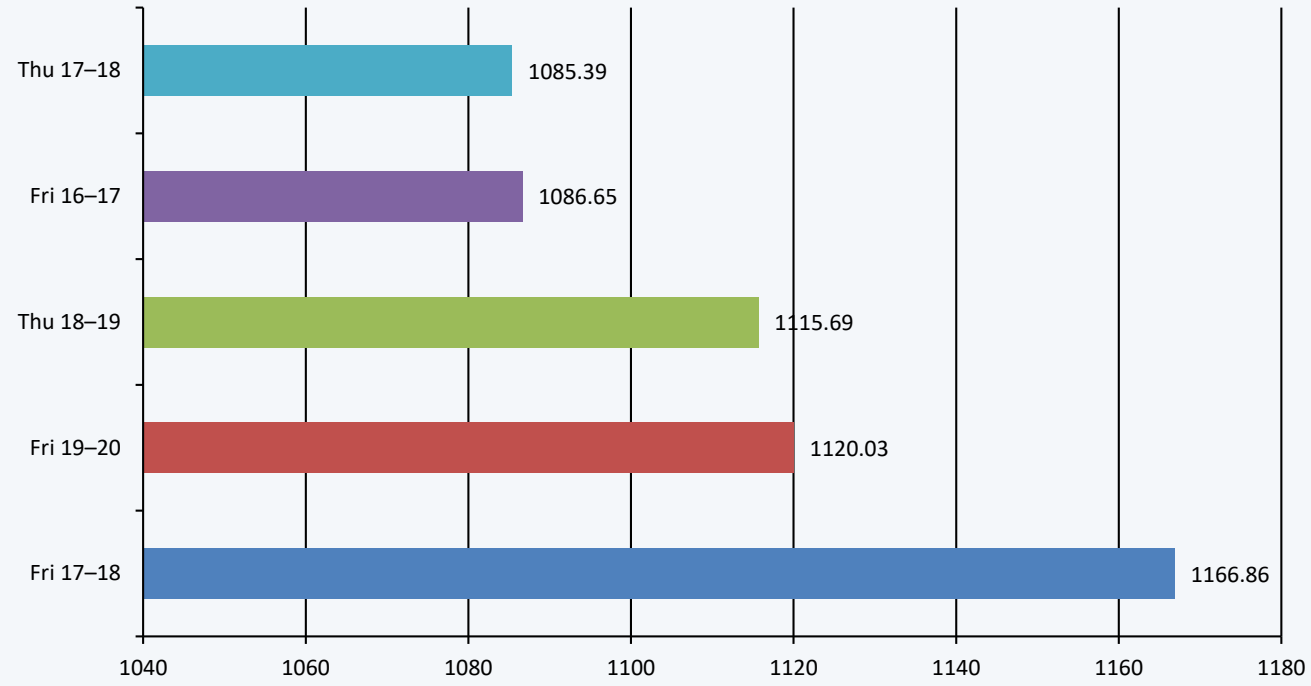
KEY TAKEAWAY

Friday 17:00–18:00 is the top listed slot, averaging 1,166.86 trips. Friday late afternoon and early evening recur in the top five rows, suggesting a useful window to test targeted driver availability.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 03 | Peak Day & Time Analysis

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Friday 5–6 PM has the highest average hourly volume in the displayed top rows.

WHAT THE RESULT TELLS US

- Friday 17:00–18:00 leads at 1,166.86 average trips.
- Late afternoon and early evening appear repeatedly in the top five.
- Consider testing driver availability around these periods, then validate with seasonal and weekday controls.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 04 | Borough Trip Distribution

CALCULATE THE PERCENTAGE OF TRIPS STARTING IN EACH PICKUP BOROUGH.

BUSINESS QUESTION 04 • SQL QUERY

```sql
# 4. Borough Trip Distribution: Find the percentage of trips starting in each pickup_borough.
with all_trip as (
select PULocationID from stg_uber_2017
union all
select PULocationID from stg_uber_2018
union all
select PULocationID from stg_uber_2017
union all
select PULocationID from stg_uber_2019
union all
select PULocationID from stg_uber_2020
), Borough_trip as (select z.Borough as pickup_Borough , count(*) as Total_trips from all_trip t join taxi_zones z on t.PULocationID=z.LocationID
group by z.Borough
)SELECT pickup_borough,total_trips, ROUND(
        total_trips * 100.0 / SUM(total_trips) OVER (),
        2
    ) AS trip_percentage
FROM borough_trip
ORDER BY trip_percentage DESC;
```

Activate Windows

Question 04 | Result & interpretation

BOROUGH TRIP DISTRIBUTION • OUTPUT PROVIDED IN YOUR SCREENSHOTS


Result Grid | Filter Rows: | Export:

pickup_borough	total_trips	trip_percentage
Manhattan	6241731	33.40
Brooklyn	5784791	30.96
Queens	5404927	28.93
Bronx	1210186	6.48
Unknown	36326	0.19

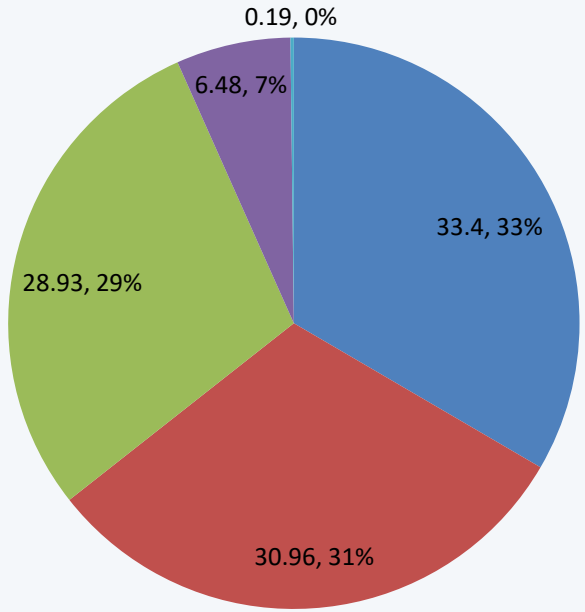
KEY TAKEAWAY

The result lists Manhattan at 33.40%, Brooklyn at 30.96%, Queens at 28.93%, the Bronx at 6.48%, and Unknown at 0.19%. The query screenshot appears to repeat the 2017 table and omit 2018 in its UNION list; correct this before final reporting.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 04 | Borough Trip Distribution

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Manhattan, Brooklyn and Queens account for the vast majority of pickup trips.

WHAT THE RESULT TELLS US

- Manhattan has the largest pickup share at 33.40%.
- Brooklyn and Queens follow closely at 30.96% and 28.93%.
- Together, these three boroughs represent about 93.3% of trips in the supplied result.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 05 | Top Pickup–Drop-off Routes

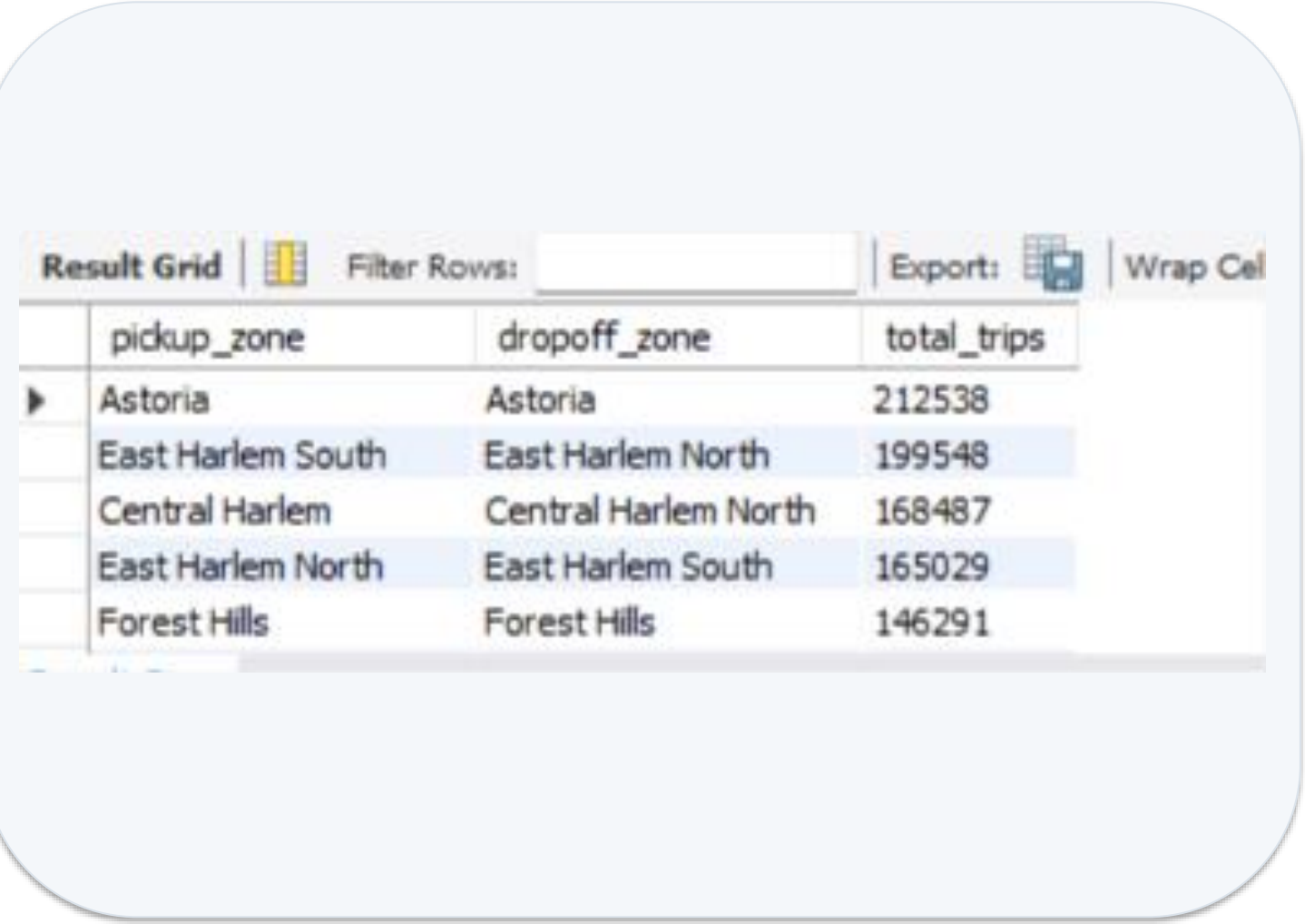
FIND THE 10 MOST FREQUENT PICKUP-ZONE TO DROP-OFF-ZONE COMBINATIONS.

BUSINESS QUESTION 05 • SQL QUERY

```sql
# 5. Top Pickup-Dropoff Routes: Identify the top 10 most frequent pickup_zone to dropoff_zone combinations.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID
    FROM stg_uber_2017
    UNION ALL
    SELECT PULocationID, DOLocationID
    FROM stg_uber_2018
    UNION ALL
    SELECT PULocationID, DOLocationID
    FROM stg_uber_2019
    UNION ALL
    SELECT PULocationID, DOLocationID
    FROM stg_uber_2020
)SELECT pu.Zone AS pickup_zone, doff.Zone AS dropoff_zone,
COUNT(*) AS total_trips FROM all_trips t
JOIN `taxi_zones` pu ON t.PULocationID = pu.LocationID
JOIN `taxi_zones` doff
    ON t.DOLocationID = doff.LocationID
GROUP BY
    pu.Zone,
    doff.Zone
ORDER BY
    total_trips DESC
LIMIT 10;
```

Question 05 | Result & interpretation

TOP PICKUP–DROP-OFF ROUTES • OUTPUT PROVIDED IN YOUR SCREENSHOTS


Result Grid | Filter Rows: | Export: | Wrap Cel

pickup_zone	dropoff_zone	total_trips
Astoria	Astoria	212538
East Harlem South	East Harlem North	199548
Central Harlem	Central Harlem North	168487
East Harlem North	East Harlem South	165029
Forest Hills	Forest Hills	146291

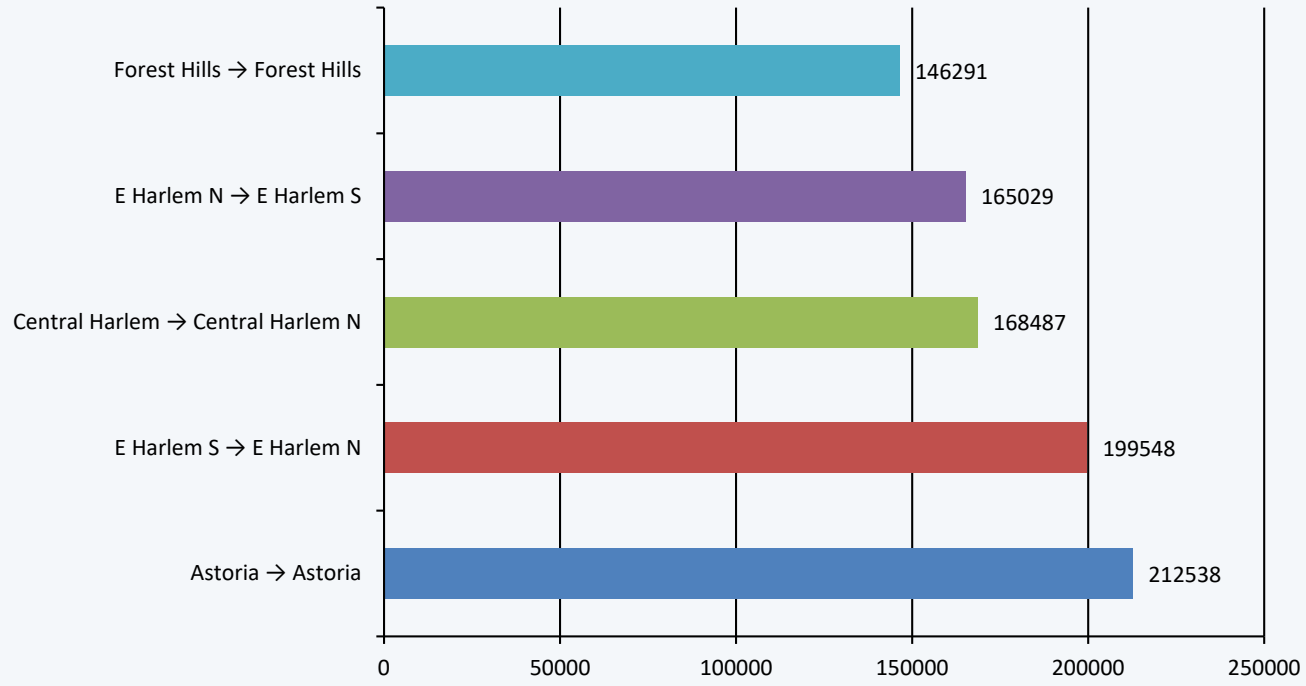
KEY TAKEAWAY

Astoria → Astoria is the most frequent displayed route with 212,538 trips. Other high-volume pairs include East Harlem South → East Harlem North (199,548) and Central Harlem → Central Harlem North (168,487). The supplied result screenshot visibly shows five rows; add the remaining rows if available.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 05 | Top Pickup–Drop-off Routes

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Short intra-neighborhood routes dominate the visible top route combinations.

WHAT THE RESULT TELLS US

- Astoria → Astoria leads with 212,538 trips.
- East Harlem South → East Harlem North is second at 199,548.
- Use zone-level drill-down to support local fleet positioning; screenshot shows five rows although query requests ten.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 06 | Passenger Load Patterns

CALCULATE AVERAGE PASSENGER COUNT FOR TRIPS BY PICKUP BOROUGH.

BUSINESS QUESTION 06 • SQL QUERY

```sql
#6. Passenger Load Patterns: Find the average passenger_count for trips in each borough.
with all_trips as ( select passenger_count,PULocationID from stg_uber_2017
union all
select passenger_count, PULocationID from stg_uber_2018
union all
select passenger_count, PULocationID from stg_uber_2019
union all
select passenger_count, PULocationID from stg_uber_2020
)
select  z.Borough, avg(t.passenger_count) as avg_passengers from all_trips t
join taxi_zones z
on t.PULocationID = z.LocationID
group by Borough;
```

Question 06 | Result & interpretation

PASSENGER LOAD PATTERNS • OUTPUT PROVIDED IN YOUR SCREENSHOTS

Borough	avg_passengers
Manhattan	1.2886551915471 83
Brooklyn	1.3282517457731373
Queens	1.4009973813224699
Bronx	1.2909298846621458
Unknown	1.2381328873346 51

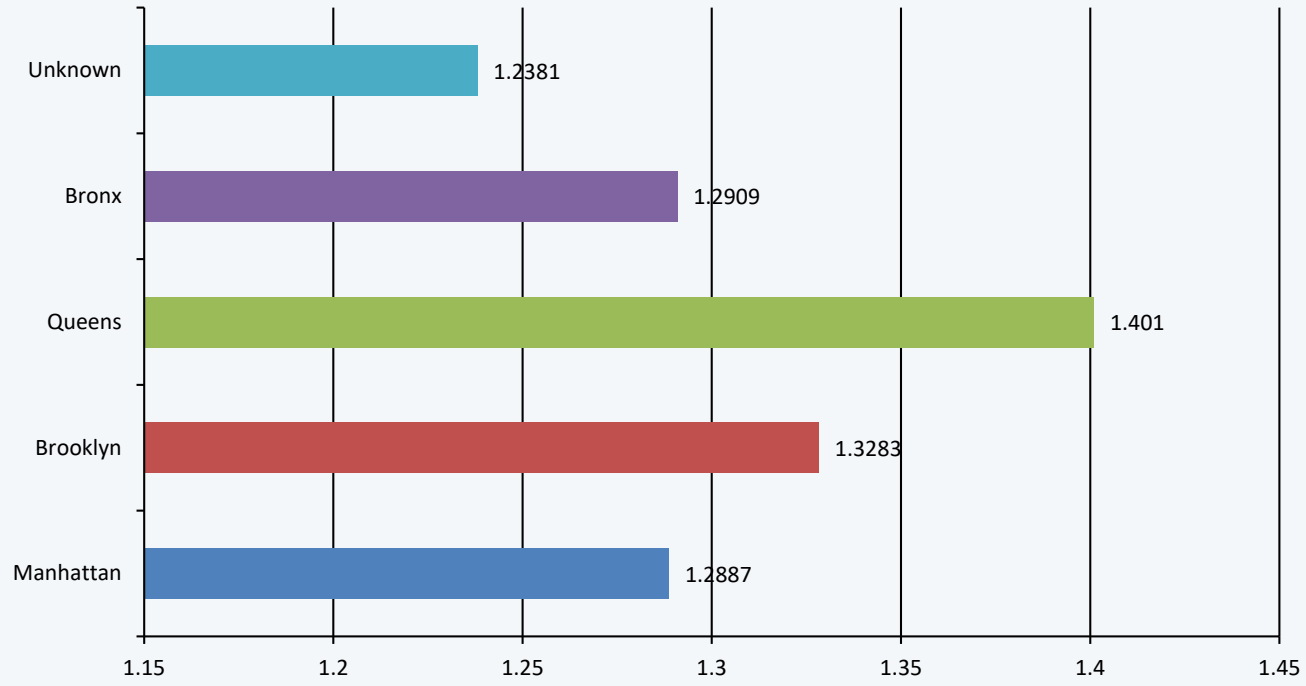
KEY TAKEAWAY

The displayed averages are approximately 1.29 passengers in Manhattan, 1.33 in Brooklyn, 1.40 in Queens, 1.29 in the Bronx, and 1.24 for Unknown. These borough averages are relatively close, with Queens highest in the shown output.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 06 | Passenger Load Patterns

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Average passenger counts are similar across boroughs; Queens is highest in the displayed output.

WHAT THE RESULT TELLS US

- Queens averages about 1.40 passengers per trip.
- Brooklyn is around 1.33; Manhattan and the Bronx are close to 1.29.
- These are averages, not vehicle-capacity measures; inspect passenger-count nulls and outliers.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 07 | Trip Distance Distribution

CALCULATE AVERAGE TRIP DISTANCE FOR EACH PICKUP-BOROUGH AND DROP-OFF-BOROUGH PAIR.

BUSINESS QUESTION 07 • SQL QUERY

```sql
#7.  Trip Distance Distribution: Determine the average trip_distance for each pickup_borough and dropoff_borough.
WITH all_trips AS (
SELECT PULocationID, DOLocationID, trip_distance
FROM stg_uber_2017
UNION ALL
SELECT PULocationID, DOLocationID, trip_distance
FROM stg_uber_2018
UNION ALL
SELECT PULocationID, DOLocationID, trip_distance
FROM stg_uber_2019
UNION ALL
SELECT PULocationID, DOLocationID, trip_distance
FROM stg_uber_2020
)
SELECT zp.Borough AS pickup_borough, zd.Borough AS dropoff_borough,
AVG(t.trip_distance) AS avg_distance FROM all_trips t
JOIN taxi_zones zp
    ON t.PULocationID = zp.LocationID
JOIN taxi_zones zd
    ON t.DOLocationID = zd.LocationID
GROUP BY
    zp.Borough,
    zd.Borough
ORDER BY avg_distance DESC;
```

Question 07 | Result & interpretation

TRIP DISTANCE DISTRIBUTION • OUTPUT PROVIDED IN YOUR SCREENSHOTS


Result Grid | Filter Rows: | Export: | Wrap Cell Content

pickup_borough	dropoff_borough	avg_distance
Manhattan	Staten Island	96.16493506493514
Staten Island	Bronx	31.581262626262617
Bronx	Staten Island	30.522021505376355
Staten Island	Queens	28.35581458145817
Queens	Staten Island	26.180663615560654

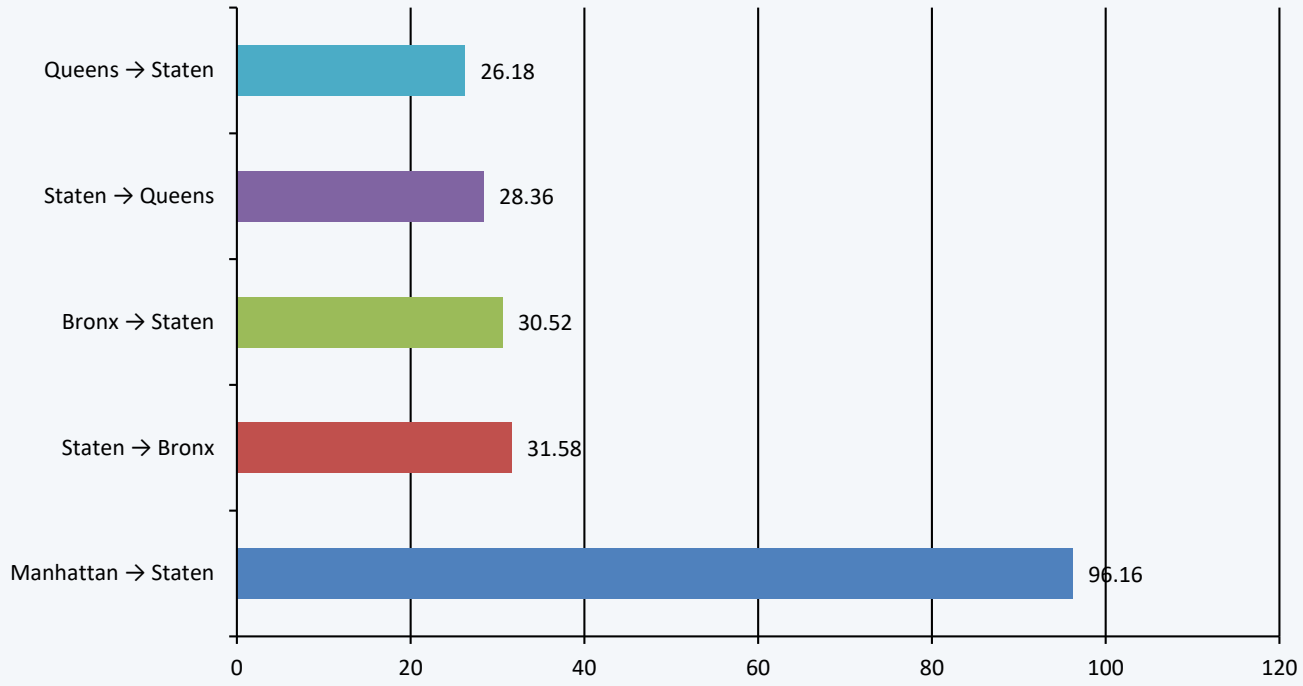
KEY TAKEAWAY

The longest average-distance pair shown is Manhattan → Staten Island at about 96.16 distance units. Staten Island → Bronx and Bronx → Staten Island are around 31.58 and 30.52. Confirm the dataset's distance unit and investigate low-frequency borough pairs.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 07 | Trip Distance by Borough Pair

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Manhattan → Staten Island is the longest average-distance pair shown.

WHAT THE RESULT TELLS US

- Manhattan → Staten Island averages about 96.16 distance units.
- Other leading pairs in the screenshot are around 26–32 units.
- Confirm the distance unit and check low-volume pairs before using these values for planning.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 08 | Payment Method Analysis

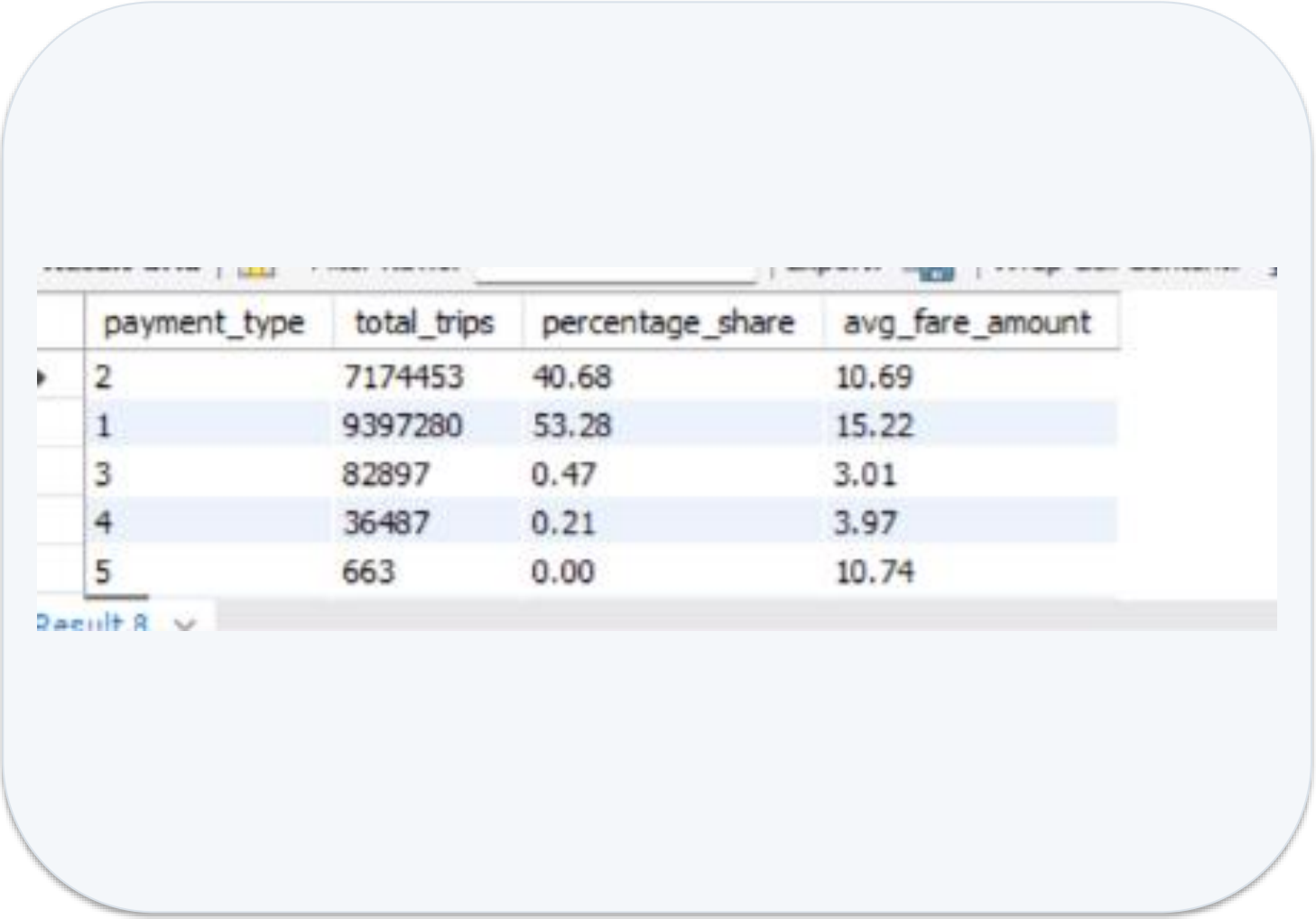
CALCULATE EACH PAYMENT TYPE'S SHARE OF TRIPS AND ITS AVERAGE FARE AMOUNT.

BUSINESS QUESTION 08 • SQL QUERY

```sql
#8. Payment Method Analysis: Calculate the percentage share of each payment_type and their average fare_amount.
WITH all_trips AS (
SELECT payment_type, fare_amount
FROM stg_uber_2017
UNION ALL
SELECT payment_type, fare_amount
FROM stg_uber_2018
UNION ALL
SELECT payment_type, fare_amount FROM stg_uber_2019
UNION ALL
SELECT payment_type, fare_amount
FROM stg_uber_2020
)SELECT payment_type,
COUNT(*) AS total_trips,ROUND( COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (),
        2
    ) AS percentage_share,
    ROUND(AVG(fare_amount), 2) AS avg_fare_amount
FROM all_trips
GROUP BY payment_type;
```

Question 08 | Result & interpretation

PAYMENT METHOD ANALYSIS • OUTPUT PROVIDED IN YOUR SCREENSHOTS



payment_type	total_trips	percentage_share	avg_fare_amount
2	7174453	40.68	10.69
1	9397280	53.28	15.22
3	82897	0.47	3.01
4	36487	0.21	3.97
5	663	0.00	10.74

Result 8

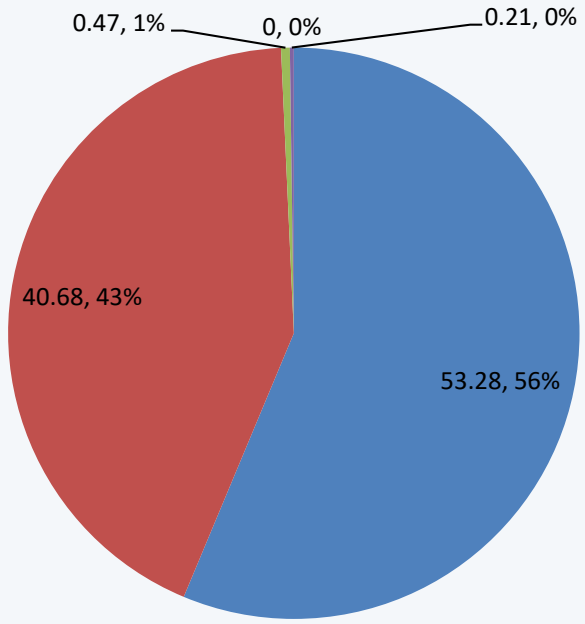
KEY TAKEAWAY

Payment type 1 accounts for 53.28% of trips with an average fare of 15.22; type 2 accounts for 40.68% with an average fare of 10.69. Types 3 and 4 are below 1% each. Decode numeric payment types into labels for a stakeholder-ready dashboard.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 08 | Payment Method Analysis

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Payment types 1 and 2 account for almost all trips in the output.

WHAT THE RESULT TELLS US

- Type 1 leads with 53.28% share and average fare 15.22.
- Type 2 contributes 40.68% with average fare 10.69.
- Translate numeric payment codes into readable labels before presenting to business stakeholders.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 09 | High-Tip Routes

FIND THE FIVE PICKUP–DROP-OFF COMBINATIONS WITH THE HIGHEST AVERAGE TIP AMOUNT.

BUSINESS QUESTION 09 • SQL QUERY

```sql
# 9. High-Tip Routes: Find the top 5 pickup-dropoff combinations with the highest average tip_amount.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID, tip_amount
    FROM stg_uber_2017
    UNION ALL
SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2018
UNION ALL
SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2019
UNION ALL
SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2020
)
SELECT zp.Zone AS pickup_zone, zd.Zone AS dropoff_zone, COUNT(*) AS total_trips,
ROUND(AVG(t.tip_amount), 2) AS avg_tip
FROM all_trips t JOIN taxi_zones zp
ON t.PULocationID = zp.LocationID
JOIN taxi_zones zd ON t.DOLocationID = zd.LocationID
WHERE t.tip_amount IS NOT NULL GROUP BY zp.Zone, zd.Zone
HAVING COUNT(*) >= 10 ORDER BY avg_tip DESC
LIMIT 5;
```

Question 09 | Result & interpretation

HIGH-TIP ROUTES • OUTPUT PROVIDED IN YOUR SCREENSHOTS

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pickup_zone	dropoff_zone	total_trips	avg_tip
Washington Heights South	Heartland Village/Todt Hill		
Morningside Heights	Eltingville/Annadale/Prince's Bay	11	17.71
LaGuardia Airport	DUMBO/Vinegar Hill	13	17.67
Flatbush/Ditmas Park	Newark Airport	10	17.51
Flushing	Newark Airport	10	17.39

Heartland Village/Todt Hill

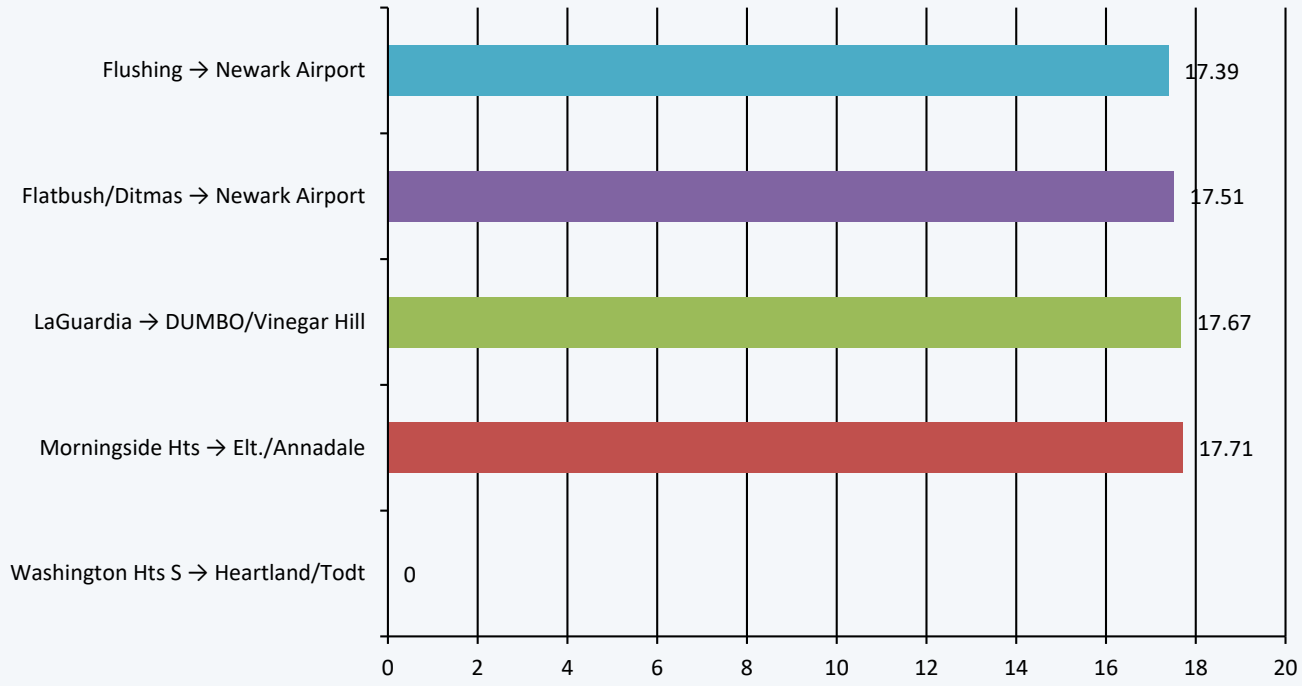
KEY TAKEAWAY

The top displayed route is Washington Heights South → Heartland Village/Todt Hill, but the screenshot's result row appears visually truncated or misaligned. Other listed routes show average tips around 17.39–17.71. Validate zone mapping and small sample sizes before interpreting rankings.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 09 | High-Tip Routes

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



The displayed top routes have high average tips but several have very small sample sizes.

WHAT THE RESULT TELLS US

- The visible result lists average tips around 17.39–17.71 for four readable rows.
- Some listed routes have only 10–13 trips, so a few transactions can shift the average substantially.
- Validate the first row's alignment and use a minimum sample-size threshold before drawing conclusions.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Question 10 | Revenue by Borough Pairs

CALCULATE TOTAL REVENUE FOR EACH PICKUP-BOROUGH TO DROP-OFF-BOROUGH PAIR.

BUSINESS QUESTION 10 • SQL QUERY

```sql
#10.  Revenue by Borough Pairs: Calculate total_amount earned for each pickup_borough to dropoff_borough pair.
WITH all_trips AS (
SELECT PULocationID, DOLocationID, total_amount
FROM stg_uber_2017
UNION ALL
SELECT PULocationID, DOLocationID, total_amount
FROM stg_uber_2018
UNION ALL
SELECT PULocationID, DOLocationID, total_amount
FROM stg_uber_2019
UNION ALL
SELECT PULocationID, DOLocationID, total_amount
FROM stg_uber_2020
)SELECT
zp.Borough AS pickup_borough,
zd.Borough AS dropoff_borough,
COUNT(*) AS total_trips,ROUND(SUM(t.total_amount), 2) AS total_revenue
FROM all_trips t JOIN taxi_zones zp ON t.PULocationID = zp.LocationID
JOIN taxi_zones zd
ON t.DOLocationID = zd.LocationID
WHERE t.total_amount IS NOT NULL
GROUP BY
    zp.Borough,
    zd.Borough
ORDER BY total_revenue DESC;
```

Question 10 | Result & interpretation

REVENUE BY BOROUGH PAIRS • OUTPUT PROVIDED IN YOUR SCREENSHOTS

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pickup_borough	dropoff_borough	total_trips	total_revenue
Manhattan	Manhattan	5220727	65094919.3
Brooklyn	Brooklyn	4273039	62748454.98
Queens	Queens	4465151	57802084.18
Brooklyn	Manhattan	774284	22919660.33
Bronx	Bronx	807010	12735130.67

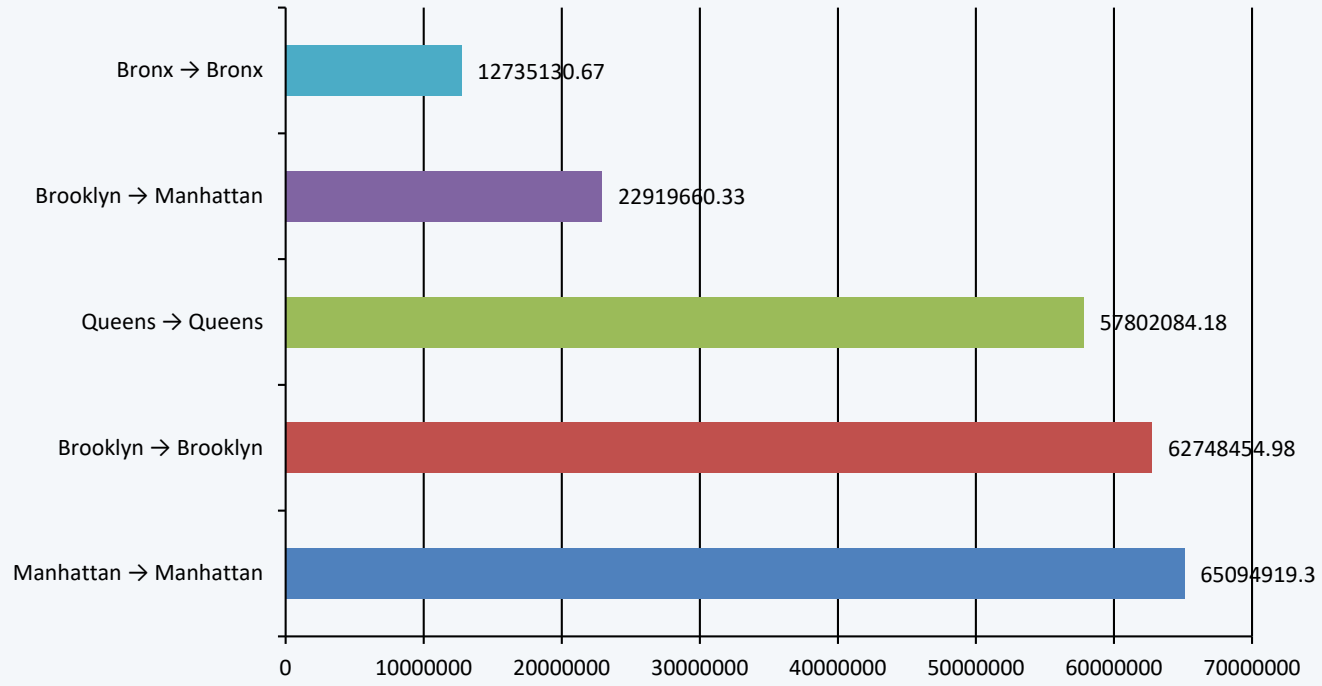
KEY TAKEAWAY

Manhattan → Manhattan has the highest displayed revenue (about 65.09M), followed by Brooklyn → Brooklyn (about 62.75M) and Queens → Queens (about 57.80M). Same-borough pairs dominate the visible top rows; check currency, null handling and data coverage.

Use this result with the Power BI dashboard for drill-down analysis.

INSIGHT 10 | Revenue by Borough Pairs

RESULT EXPLAINED • VISUAL SUMMARY • BUSINESS IMPLICATION



Same-borough trips dominate the top revenue rows in the supplied output.

WHAT THE RESULT TELLS US

- Manhattan → Manhattan leads at about 65.09M in total revenue.
- Brooklyn → Brooklyn and Queens → Queens follow at about 62.75M and 57.80M.
- Confirm currency/scale and date coverage before comparing revenue across years or borough pairs.

Decision cue: use this insight alongside date, borough and route filters in the Power BI dashboard.

Executive summary & validation checklist

KEY FINDINGS FROM THE TEN SUPPLIED SQL RESULT SETS

Trip volume	2018 is the highest-count year in the supplied output; subsequent years fall sharply.
Time	Friday 5–6 PM is the busiest listed hour/day slot.
Geography	Manhattan, Brooklyn and Queens dominate pickup share; local borough pairs also lead displayed revenue.
Passengers & distance	Queens has the highest displayed average passenger count; Manhattan → Staten Island has the longest displayed average distance.
Payments & tips	Payment types 1 and 2 dominate the result; high-tip route ranking needs sample-size and mapping checks.

Before final publication: fix the borough UNION list, verify year coverage/revenue units, label payment types, and validate tip-route row alignment.

Next stage | Power BI & predictive analytics

THIS SQL DECK IS DESIGNED TO MERGE WITH THE MATCHING DASHBOARD PRESENTATION LATER.

1	**Power BI**	Add dashboard overview, time demand, geographic patterns, passenger/distance, payment/tips pages.
2	**Demand forecasting**	Add model features, train/test split, evaluation metrics, and forecast chart.
3	**High-tip prediction**	Add target definition, model selection, evaluation metrics, and practical caveats.
4	**Final recommendation**	Combine SQL and dashboard evidence into operational actions.

End of SQL analysis section • Power BI screenshots to follow

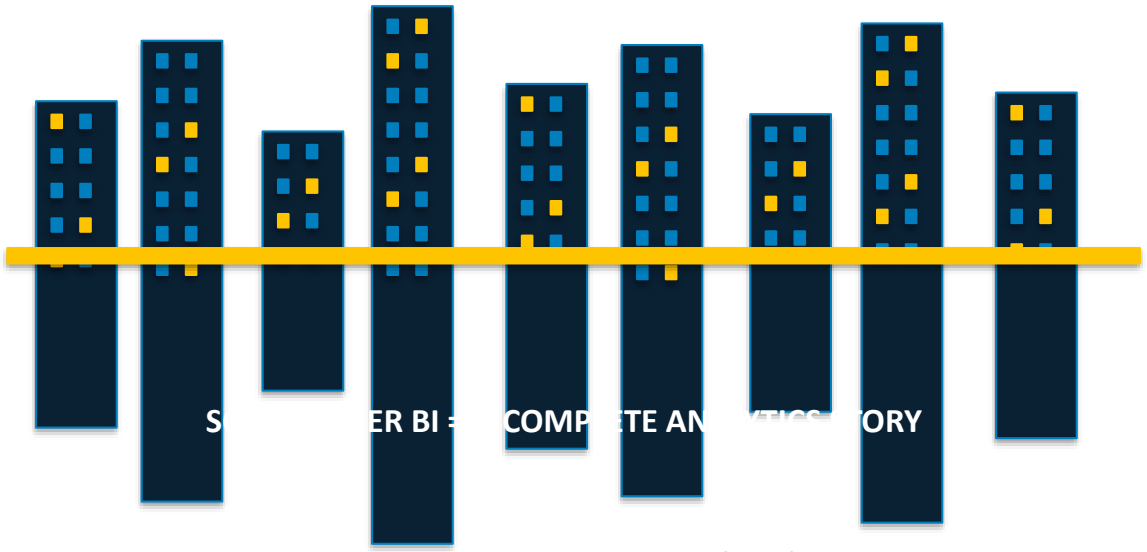
UBER TRIPS ANALYTICS

POWER BI DASHBOARD STORY

From operational data to visual insights and decisions

POWER BI EDITION • 5 DASHBOARDS

NYC GREEN TAXI TRIPS | 2017–2020



Five interactive report pages explained

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TRIPS ANALYTICS

Presentation Roadmap

POWER BI EDITION

WHAT THIS POWER BI EDITION COVERS

01	**Trip & Revenue Overview**	Scale, revenue, fare levels and yearly movement
02	**Time-Based Demand**	Hourly peaks and weekday patterns
03	**Geographic Patterns**	Borough mix, zones and trip distance
04	**Payment & Tips**	Payment mix and borough-level tipping
05	**Passenger & Distance**	Passenger load and distance differences

The supplied screenshots are preserved as dashboard evidence; the following slides interpret the visible metrics.



NYC GREEN TAXI DATA PROJECT

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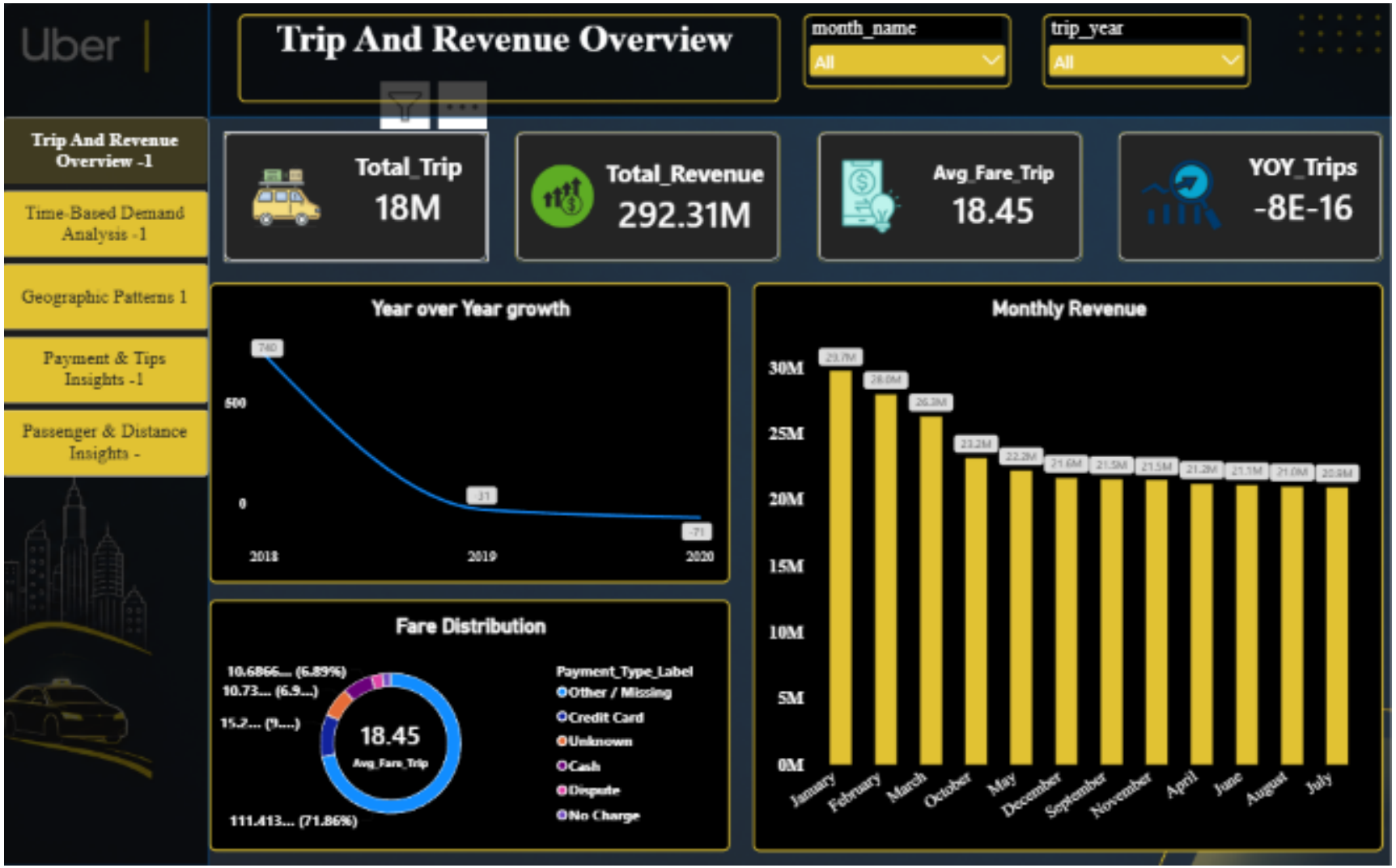
TRIPS ANALYTICS

NYC GREEN TAXI DATA PROJECT

Dashboard 1: Trip & Revenue Overview

ORIGINAL POWER BI SCREENSHOT

FILTERS AND VISUALS AS PROVIDED



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TRIPS ANALYTICS

Dashboard 1: Key Findings

INTERPRETATION • BUSINESS USE

KEY METRICS VISIBLE IN THE DASHBOARD

TOTAL TRIPS	TOTAL REVENUE	AVG FARE / TRIP	YOY TRIPS
18M	292.31M	18.45	-8E-16

WHAT THE DASHBOARD SHOWS

- The page summarizes the overall trip scale, total revenue and average fare.
- The monthly revenue chart is relatively high early in the visible period and trends lower later.
- The yearly growth line falls sharply after 2018; validate year coverage and measure formatting before interpreting the KPI.

HOW THE BUSINESS CAN USE IT

- Use monthly and yearly filters to isolate periods with material demand changes.
- Compare revenue and trip counts together: revenue can change because of trip volume, fare mix or data coverage.
- Fix the YoY KPI format; -8E-16 is effectively zero due to floating-point/measure formatting, not a meaningful business result.

Interpretation is based on the values visible in the supplied screenshot; validate measures and filters in the PBIX before operational use.

NYC GREEN TAXI DATA PROJECT

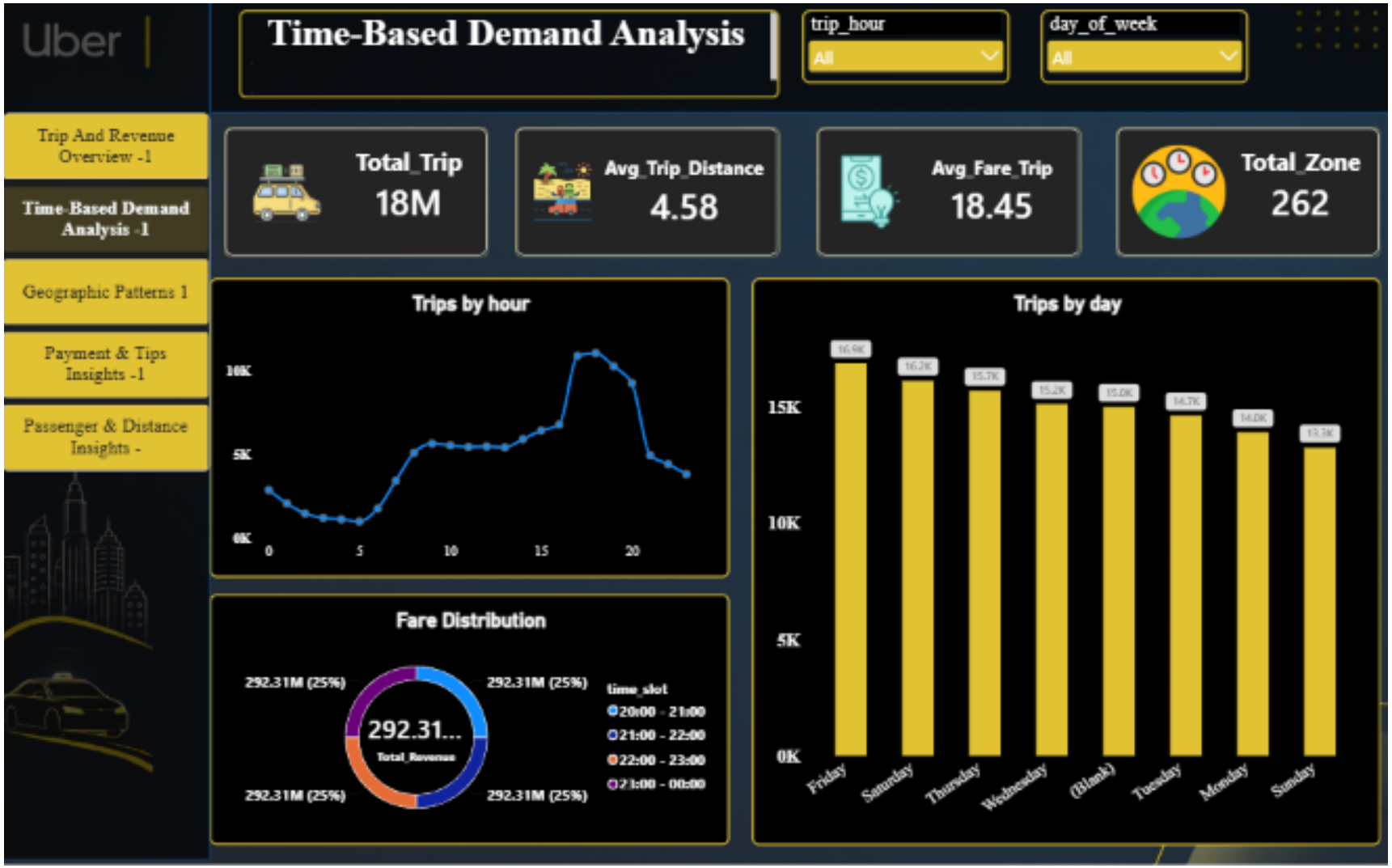
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TRIPS ANALYTICS

Dashboard 2: Time-Based Demand Analysis

ORIGINAL POWER BI SCREENSHOT

FILTERS AND VISUALS AS PROVIDED



NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

Dashboard 2: Key Findings

INTERPRETATION • BUSINESS USE

KEY METRICS VISIBLE IN THE DASHBOARD

TOTAL TRIPS	AVG TRIP DISTANCE	AVG FARE / TRIP	TOTAL ZONES
18M	4.58	18.45	262

WHAT THE DASHBOARD SHOWS

- The hourly line shows demand rising through the day, with the strongest activity around late afternoon / early evening.
- The weekday bar chart ranks Friday highest among the displayed days, followed by Saturday and Thursday.
- The report provides hour and day filters to examine demand at more granular levels.

HOW THE BUSINESS CAN USE IT

- Test driver positioning before the late-afternoon/evening peak, using more than one representative week.
- Compare weekdays with weekends and seasonality before changing shift plans.
- Confirm the donut visual is mapped to the intended measure: its labels currently repeat total revenue rather than a time-slot distribution.

Interpretation is based on the values visible in the supplied screenshot; validate measures and filters in the PBIX before operational use.

NYC GREEN TAXI DATA PROJECT

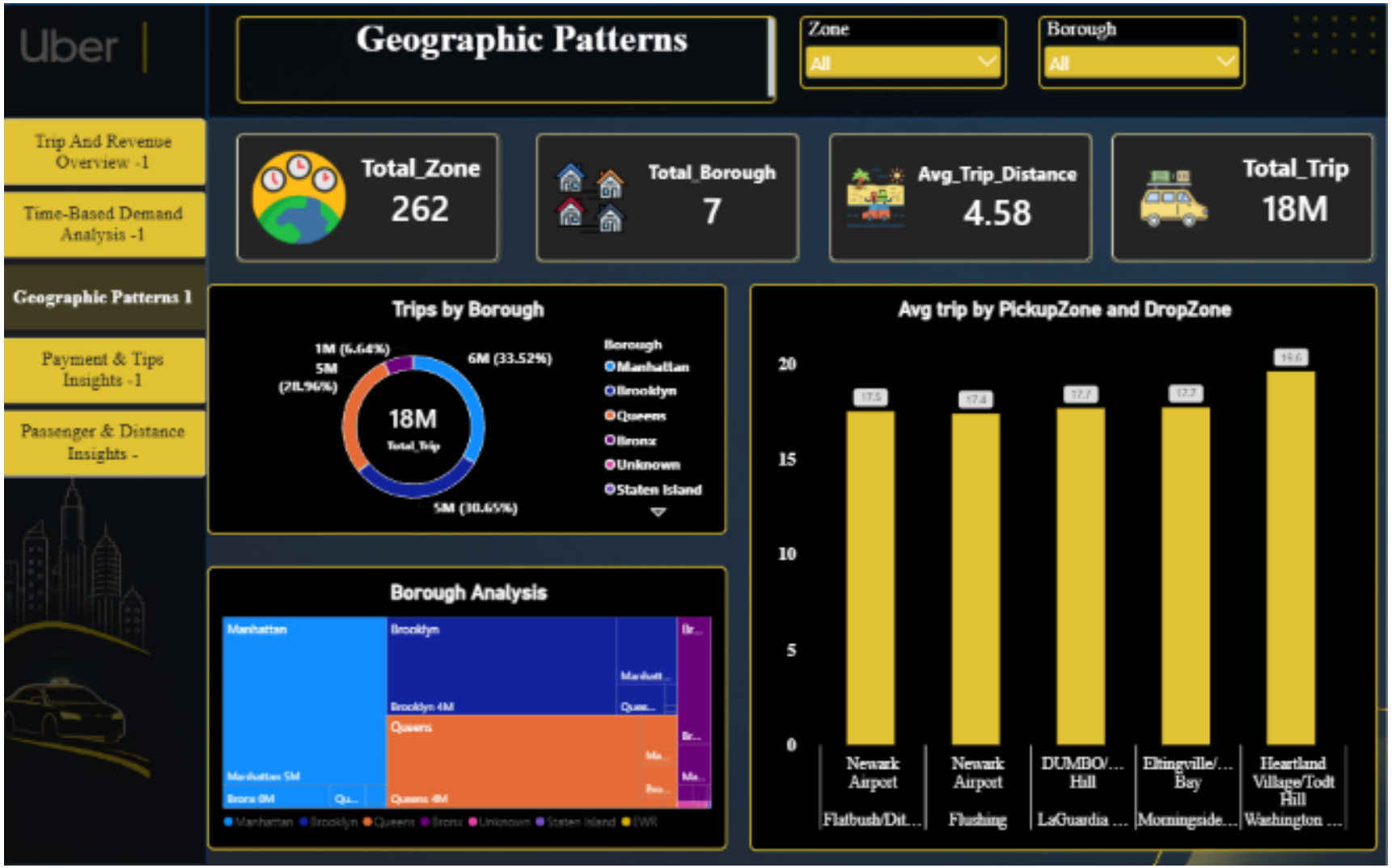
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TRIPS ANALYTICS

Dashboard 3: Geographic Patterns

ORIGINAL POWER BI SCREENSHOT

FILTERS AND VISUALS AS PROVIDED



NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

Dashboard 3: Key Findings

INTERPRETATION • BUSINESS USE

KEY METRICS VISIBLE IN THE DASHBOARD

TOTAL ZONES	TOTAL BOROUGHS	AVG TRIP DISTANCE	TOTAL TRIPS
262	7	4.58	18M

WHAT THE DASHBOARD SHOWS

- The borough donut shows Manhattan as the largest share, with Brooklyn and Queens also contributing substantial volume.
- The treemap breaks activity down into borough and zone groups for drill-down exploration.
- The pickup-zone/drop-zone chart highlights several route averages, with the highest displayed value around 19.6.

HOW THE BUSINESS CAN USE IT

- Prioritize zone-level monitoring in high-volume boroughs, while checking coverage in lower-volume areas.
- Use the route visual to identify where short-distance and longer-distance patterns differ.
- Validate zone/borough labels and route sample sizes before dispatch or pricing decisions; a high average alone does not imply high demand.

Interpretation is based on the values visible in the supplied screenshot; validate measures and filters in the PBIX before operational use.

NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

NYC GREEN TAXI DATA PROJECT

Dashboard 4: Payment & Tips Insights

ORIGINAL POWER BI SCREENSHOT

FILTERS AND VISUALS AS PROVIDED



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TRIPS ANALYTICS

NYC GREEN TAXI DATA PROJECT

Dashboard 4: Key Findings

INTERPRETATION • BUSINESS USE

KEY METRICS VISIBLE IN THE DASHBOARD

AVERAGE TIP	PAYMENT TYPE SHARE
1.04	1.00

WHAT THE DASHBOARD SHOWS

- The payment distribution is led by Credit Card at about 60.11% in the visible chart.
- Cash contributes about 29.38%, while Other / Missing is around 10.36%.
- The borough tip chart shows EWR far above the other displayed categories, at approximately 11.8; remaining categories are much lower.

HOW THE BUSINESS CAN USE IT

- Support card-payment reliability while keeping cash workflows available for riders who use it.
- Investigate Other / Missing records to improve payment reporting completeness.
- Check EWR sample volume, tip outliers and the borough-versus-location classification before interpreting it as a typical borough pattern.

Interpretation is based on the values visible in the supplied screenshot; validate measures and filters in the PBIX before operational use.

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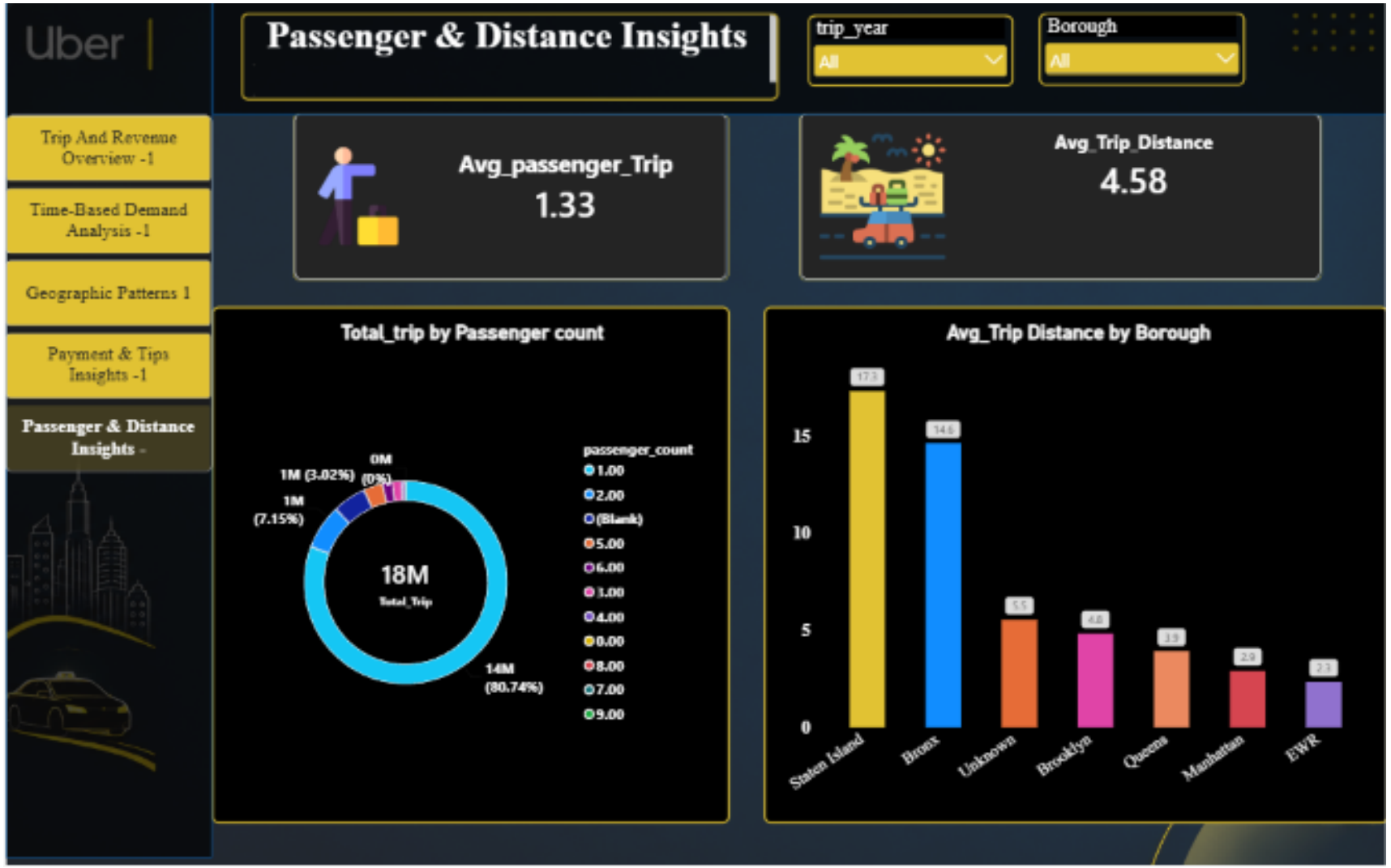
TRIPS ANALYTICS

NYC GREEN TAXI DATA PROJECT

Dashboard 5: Passenger & Distance Insights

ORIGINAL POWER BI SCREENSHOT

FILTERS AND VISUALS AS PROVIDED



Uber

TRIPS ANALYTICS

Dashboard 5: Key Findings

INTERPRETATION • BUSINESS USE

KEY METRICS VISIBLE IN THE DASHBOARD

AVG PASSENGERS / TRIP	AVG TRIP DISTANCE
1.33	4.58

WHAT THE DASHBOARD SHOWS

- The passenger-count donut is dominated by one-passenger trips (about 80.74% shown).
- The average trip distance differs by borough: Staten Island is highest in the displayed chart, followed by the Bronx.
- The dashboard combines passenger-load distribution with a borough comparison of average distance.

HOW THE BUSINESS CAN USE IT

- Use the high share of single-passenger trips when planning capacity, but do not assume every trip needs a larger vehicle.
- Review long-distance borough patterns for suitable vehicle positioning and service coverage.
- Validate blank/zero passenger counts and confirm distance units before using the chart for operational targets.

Interpretation is based on the values visible in the supplied screenshot; validate measures and filters in the PBIX before operational use.

NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

Problems This Project Helps Identify

FROM ANALYTICS TO ACTION

Demand imbalance	Which years, weekdays and hours have higher or falling trip activity?
Revenue volatility	Are revenue changes driven by trip volume, fare levels or data completeness?
Geographic coverage gaps	Which boroughs/zones dominate activity, and where might fleet positioning need review?
Route efficiency	Which pickup–drop-off pairs are common, unusually long, or operationally important?
Payment and tip issues	Which payment types dominate, where are missing categories, and which tip averages need validation?
Passenger and distance patterns	How often are trips single-passenger, and which boroughs show longer average journeys?

SQL quantifies the questions; Power BI makes patterns and exceptions easier to explore.



NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

Data Quality & Interpretation Checks

IMPORTANT BEFORE FINAL SUBMISSION

- The SQL results and dashboard cards show different total-trip figures in places; reconcile date range, table coverage and filters.
- The YoY Trips card displays -8E-16, which is a formatting / floating-point artifact and should not be presented as a meaningful percentage.
- Some visuals appear to use mismatched measures or labels (for example, repeated total-revenue labels in a time-slot donut).
- Confirm currency, distance units, missing values, payment code mappings and passenger-count outliers.
- For route and tip analysis, include minimum trip counts and inspect outliers before making recommendations.

A transparent validation slide strengthens the credibility of the capstone.

NYC GREEN TAXI DATA PROJECT

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TRIPS ANALYTICS

Final Conclusion: SQL + Power BI

CAPSTONE PROJECT • END-TO-END VALUE

SQL provides the evidence. Power BI turns it into an operational story.

WHAT SQL CONTRIBUTES

- Combines annual trip tables and calculates trip, revenue and route metrics.
- Ranks peak periods, borough shares, frequent routes, passenger loads and tip/payment patterns.
- Makes business questions measurable and reproducible.

WHAT POWER BI CONTRIBUTES

- Shows KPIs and trends together across five dashboard pages.
- Allows interactive filtering by time, borough, zone and payment category.
- Helps stakeholders explore patterns without rerunning SQL manually.

Together, SQL and Power BI can help teams investigate demand peaks, revenue shifts, geographic imbalances, route patterns, payment-quality issues and passenger-distance trends. The results support better-informed fleet scheduling and service planning—but recommendations should be confirmed after reconciling totals, validating measures, and checking data quality.

NEXT STEP: validate the underlying model, finalize the report filters, and combine this deck with the SQL edition for submission.



NYC GREEN TAXI DATA PROJECT

UBER TRIPS ANALYTICS

SQL analysis + Power BI dashboards
From raw trip data to actionable operational insights

